

1 Intensity Transformation (40pts)

a) (10pts) Do you think this is a good image or not? Explain why. According to what you have learned in intensity transformation, please propose two solutions to enhance this image (image included in ./code_data/test1.png).



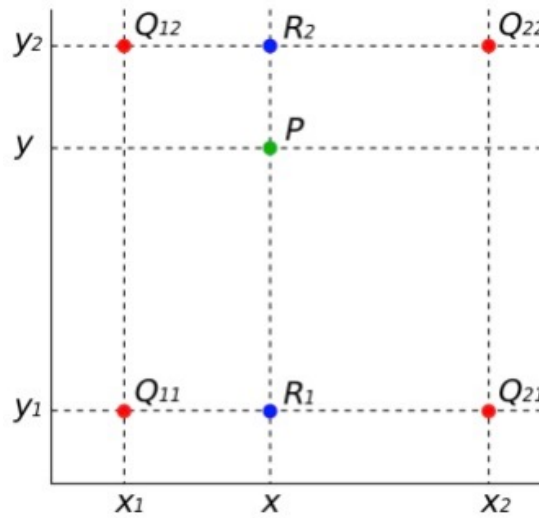
Please use the provided skeleton code ("./code_data/assignment1_1.py") for the following questions. You need to write code in specific positions.

- b)** (10pts) Please refer to the histogram of this image (Line 30 and refer to Line 11 - 20), and describe it in your report.
- c)** (10pts) Call the function ("cv2.equalizeHist()") in opencv to realize histogram equalization and provide the histogram and image after histogram equalization in your report. (You may need to refer: https://docs.opencv.org/3.4/d5/daf/tutorial_py_histogram_equalization.html.)
- d)** (15pts) Please write your own code for histogram equalization (Line 55, 66, 70) and apply your own code to process the image and include the results in your report.
- e)** (15pts) Analyze the differences before and after histogram equalization.

2 Image Interpolation (30pts)

a) (15pts) Compare the nearest interpolation and bilinear interpolation and briefly describe their advantages and disadvantages.

b) (25pts) Please use provided skeleton code (`./code.data/assignment1.2.py`) to implement bilinear interpolation (Line 36, 37, 38) and use it to resize an image to size 200×200 . As shown in the figure below, P is the target point in the source image and Q_{11} , Q_{12} , Q_{21} and Q_{22} are the corresponding nearest points in the source image. Please use these four nearest points to calculate the target point P . (Referring to the provided code, P is (src_x, src_y) , Q_{11} is (src_x-1, src_y-1) , Q_{12} is (src_x-1, src_y-2) , Q_{21} is (src_x+2, src_y-1) and Q_{22} is (src_x+2, src_y-2))



Note: we have provided the code skeleton in `assignment1.2.py`

3 Spatial Filtering and Image Sharpening (30pts)

a) (10pts) Refer to the slide Topic 3: Spatial Filtering from Page 36 to 56. Given an image with size $M \times N$ and a filter kernel with size $m \times n$, please briefly describe the process of linear spatial filtering using convolution, and show how many multiplication operations need to be performed? If the kernel is separable, how much speed boost can be obtained (considering multiplication calculations only)?

b) (20pts) Please use the provided skeleton code (`./code_data/assignment1_3.py`) to conduct image sharpening, using function **Laplacian** from OpenCV, which sharpens the edges of the image. Moreover, conduct the code with different values of magnifying factor and parameters in Laplacian Function, and analyze the results.

Note: we have provided the code skeleton in [assignment1_3.py](#)